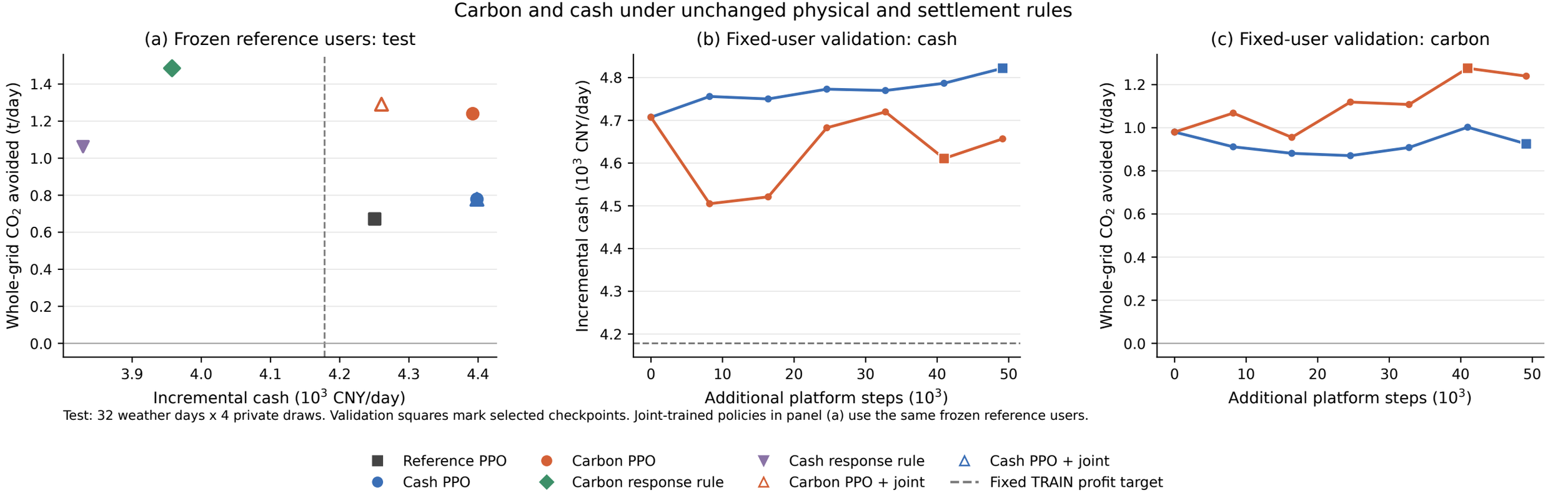
Carbon and cash under unchanged physical and settlement rules
(a) Frozen reference users: test
Whole-grid CO2 avoided (t/day)
Incremental cash (10^3 CNY/day)
(b) Fixed-user validation: cash
Incremental cash (10^3 CNY/day)
Additional platform steps (10^3)
(c) Fixed-user validation: carbon
Whole-grid CO2 avoided (t/day)
Additional platform steps (10^3)
Test: 32 weather days x 4 private draws. Validation squares mark selected checkpoints. Joint-trained policies in panel (a) use the same frozen reference users.
Reference PPO
Cash PPO
Carbon PPO
Carbon response rule
Cash response rule
Carbon PPO + joint
Cash PPO + joint
Fixed TRAIN profit target